Measuring Muscle Co-contraction During Simulated Laparoscopy

Using Surface EMG Approach

Training, Sensors & Simulation | University of Twente

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Background

The Problem with Traditional Surgical Training

Surgical training historically relied on observation and practice on real patients where any mistake could cost health or life.

Modern safety standards reject this apprenticeship model as unacceptable (Shahrezaei et al., 2024).

Simulation has emerged as the primary alternative, reducing critical errors by up to 61% (Mascheroni et al., 2024).

However, standard simulator metrics only capture the outcome of the task not how the task is performed.

61%

Fewer critical errors
with simulation training

?

No insight into HOW
the task is performed

Rationale

Why Physiological Measurement?

Outcome ≠ Process

Two participants may score equally but use entirely different motor strategies

Muscle Co-contraction

Every joint has two muscles: *Agonist* bends, *Antagonist* extends. When both activate at the same time, that is co-contraction, a sign of stiffness and reduced control

Real-time Feedback

Co-contraction percentage is displayed after each trial, enabling coaches to give targeted, personalised recommendations between trials

Research Questions & Hypotheses

Is forearm muscle co-contraction measured via surface EMG associated with laparoscopic simulation performance, and does it change across repeated trials in a way that reflects motor learning?

H_1 Participants with higher forearm muscle co-contraction will achieve lower LapSim efficiency and tissue handling scores, reflecting greater neuromuscular stiffness and less refined motor control

H_2 Across five consecutive trials, co-contraction index will decrease as LapSim performance scores improve, indicating progressive motor adaptation and increasing neuromuscular efficiency

Operationalisation: sEMG co-contraction index (% of trial time both agonist & antagonist simultaneously exceed mean baseline + 0.1 SD threshold) | LapSim efficiency & tissue handling scores

Methods

Participants & Experimental Design

Participants	3 (aged 22-25), MSc Human Factors students at University of Twente
Expertise	All novice to surgery
Design	Within-subjects: 1 block of 5 trials per participant
Baseline	10-second resting measurement before each block
Task	LapSim laparoscopic navigation: poke tissue targets with indicated instrument

LapSim

Methods

LapSim Simulator Task

Instrument Control

Scissor-grip with middle finger and thumb; index finger rotates jaw via gear mechanism

Task Procedure

Calibrate instruments → Lift to start → Poke tissue targets (yellow dots) with the indicated instrument (lit green)

Performance Metrics

Efficiency: time, path length, angular path

Tissue handling: damage amount/size, out-of-view time

Benchmark Feedback

Performance normalised against expert-level data. Feedback bar: red → orange → green scale.

Limitation: ceiling effect at 100% — progression beyond expert level not detectable from LapSim alone.

Methods

sEMG Measurement Setup

Electrode Placement

Agonist (dorsal):

Extensor digitorum - electrodes 1-3

Antagonist (ventral):

Flexor digitorum superficialis & palmaris longus — electrodes 4-6

Reference electrodes on elbow bone

Mid-belly placement, parallel to muscle fibres, ~20 mm centre-to-centre (SENIAM guidelines)

Technical Details

Hardware:

STM32 Nucleo F446ZE board (max 250 Hz)

Software:

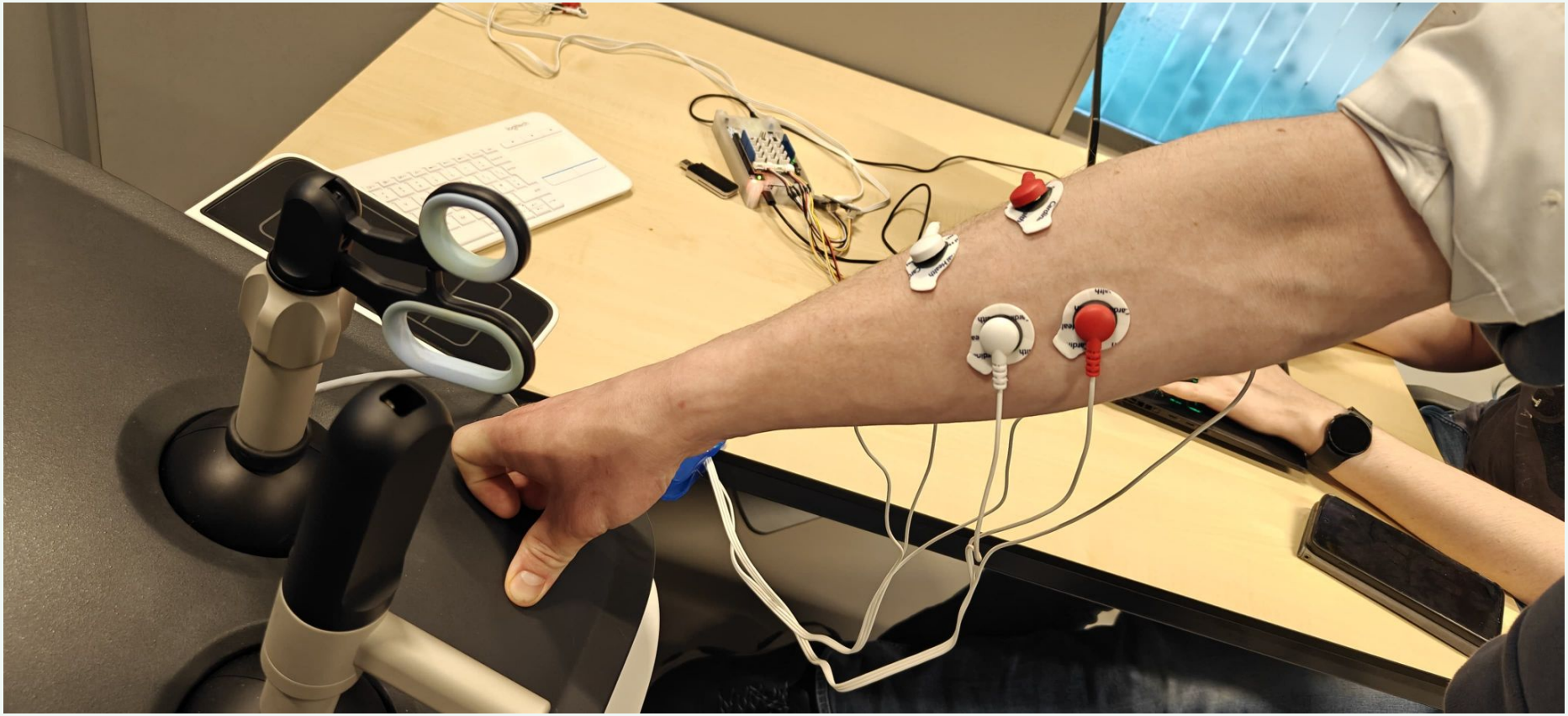
Ylab Pro + Ylab Zero (Rust-based)

Detection:

Bipolar (Single Differential) — cancels common-mode noise (e.g. PLI)

Sampling:

240 Hz (hardware-constrained)



Human Factors

Human Factors Design Considerations

Problem

Loose, dangling sEMG wires from sensor contact points to the STM32 board impede participant movement, reduce concentration, and introduce noise artefacts.

Solutions

1. Arm sleeves

Runner’s phone bracelets / armguards to hold wires against the arm

2. Elevated platform

High table at elbow height for STM32 board + laptop, minimising wire drop

3. Cable clips

Cable organisers directing wires upward, reducing movement artefacts

Methods

Co-contraction Analysis

1	**Baseline**	Calculate mean and SD of sEMG for each muscle group
2	**Threshold**	Set activation criterion: signal > mean_baseline + 0.1 × SD_baseline
3	**Co-contraction**	Identify moments when BOTH agonist and antagonist exceed threshold simultaneously
4	**Percentage**	Calculate % of trial time meeting co-contraction criteria
5	**Comparison**	Compare percentage across 5 trials to track learning progression

Methods

Feedback Visualisation System

Visual Feedback Website

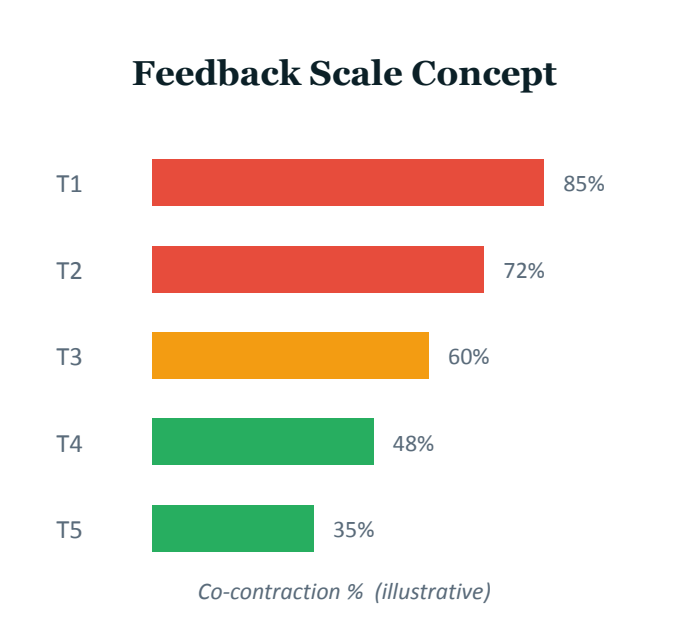
Custom-built web interface displays co-contraction percentage per trial, enabling real-time comparison across the 5-trial block.

Integrated Feedback Loop

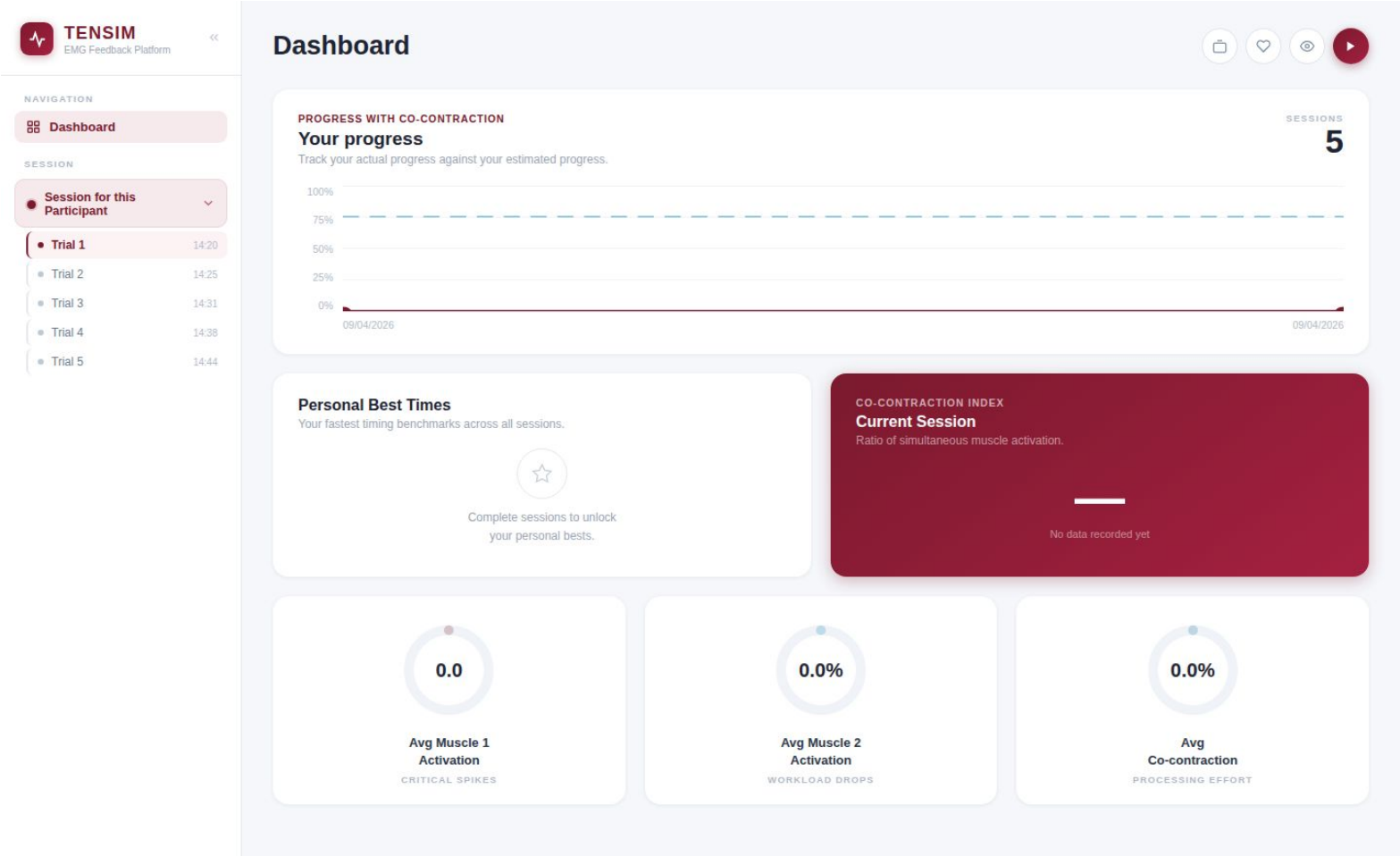
After combining sEMG and LapSim results, the participant receives personalised coaching recommendations (e.g., "try to relax for the next trial").

Goal

Provide surgical coaches with a cost-effective, quantitative tool for monitoring and improving trainee performance.

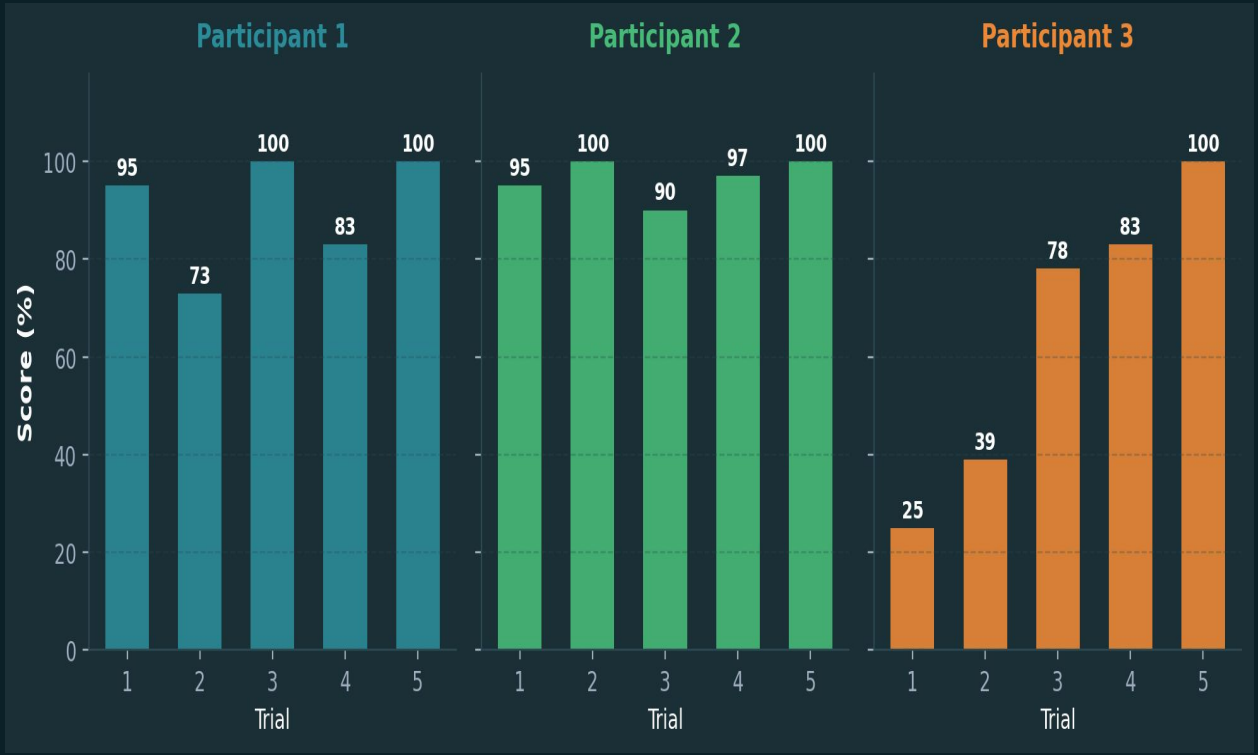


Example of providing Feedback



Results

LapSim performance scores across trials



Linear Model predicting Score through Percentage of Co-Contraction is not significant

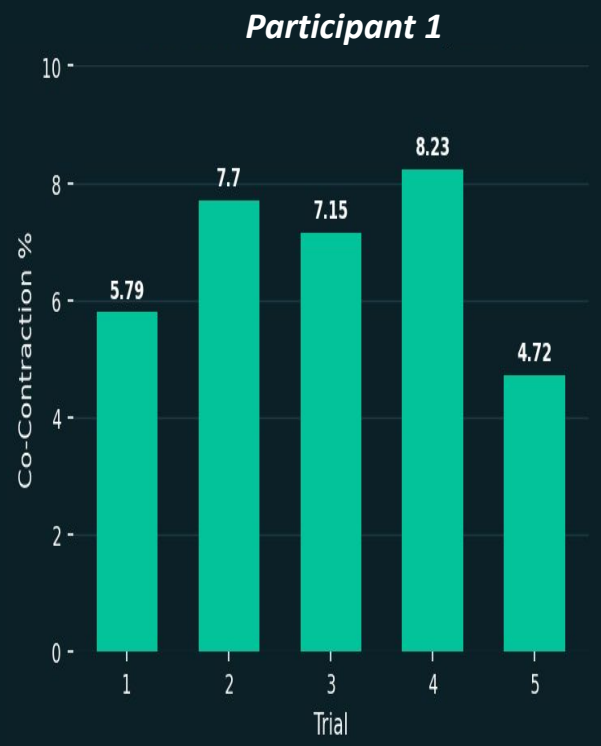
$p > 0.05$ ($p = 0.454$)

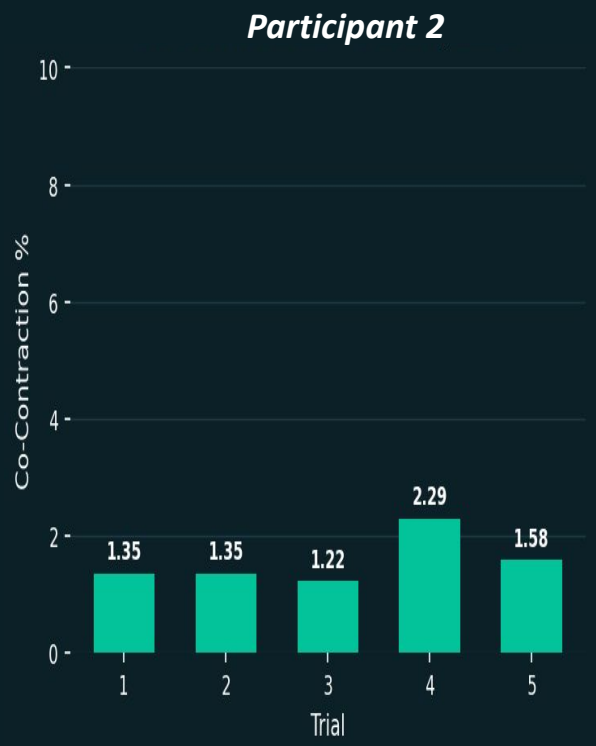
Linear Model predicting Status through Percentage of Co-Contraction is not significant

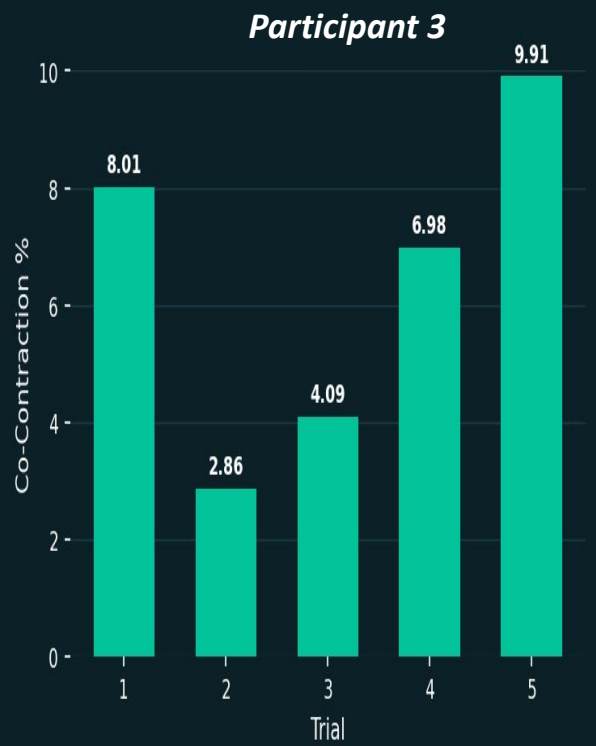
$p > 0.05$ ($p = 0.086$)

Results

Co-Contraction outcomes across trials







Results

H_1: Not Supported

Co-contraction does not significantly predict LapSim score.

$p = 0.454$, $R^2 = 0.044$

No linear relationship between CCI and task performance was found.

H_2: Not Supported

Co-contraction does not significantly predict pass/fail status.

$p = 0.086$, $R^2 = 0.210$

p-value did not fall below the 0.05 threshold.

Why?

User errors?

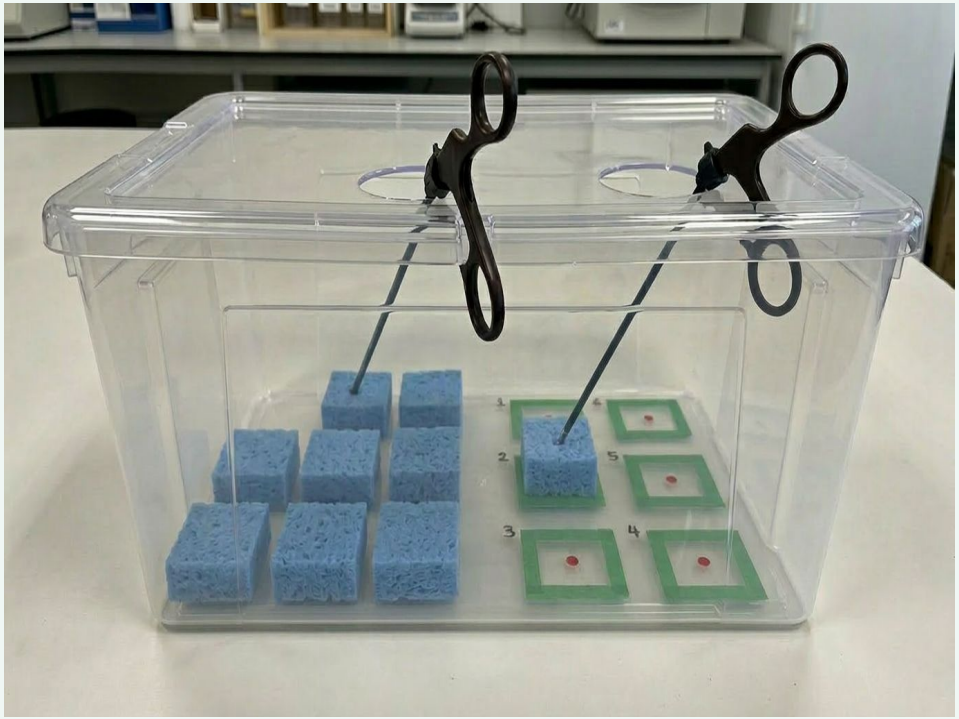
Small sample size?

Incorrect EMG analysis?

Faulty equipment?

Replication

Low-Cost Box Simulator Alternative



Based on the Inovus Medical box simulator design

- Large plastic box with two instrument holes in lid
- 6 sponge pieces (2×3 grid) on left, 6 target sections on right
- Pass items left → right instrument, place in target zone

Advantages

- Cost-effective
- Uses same sEMG/IMU setup
- Adjustable difficulty levels
- No LapSim access required
- Physical instrument handling preserved

Discussion

Key Challenges

Language Barrier

Ylab Zero written in Rust for Linux; team proficient in Python on Windows. Solved with separate Python processing pipeline.

Simulator Access

LapSim unavailable until late in project. Compensated by preparing methods and building box simulator alternative.

Manual Sync

No programmatic sync between sEMG and LapSim. Manual start ±1 sec before trial onset introduces timing imprecision.

Small Sample

3 participants. Exploratory design validates setup feasibility rather than statistical generalisability.

Future Directions

IMU Sensor & Signal Processing

IMU MotionSensor

Placed on the dorsum of the hand (centre). Measures acceleration via inertia as a proxy for movement smoothness: high magnitude = jerky; low = smooth.

sEMG Signal Pipeline

Synchronisation

sEMG manually started ~1 second before LapSim trial onset. Data output: .ylb files with 5 columns (timestamp, apparatus ID x2, electrode marker, signal value). Separate files per participant per trial.

Conclusion

This project demonstrates a cost-effective, replicable setup for measuring forearm muscle co-contraction during simulated laparoscopy using sEMG sensor.

The visual feedback system provides surgical coaches with quantitative, real-time data about trainee motor control beyond what standard simulator metrics offer.

Future work should include larger samples, automated measurement synchronisation, and integration with more advanced simulator platforms.

Thank you

Questions?